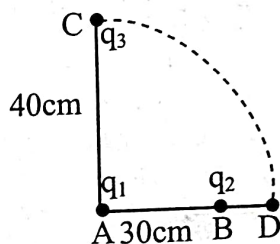


Electrostatic Potential and Capacitance

Work done to move charge in Electric Field, Electrostatic Potential Energy

Two charges q_1 and q_2 are separated by a distance of 30 cm. A third charge q_3 initially at 'C' as shown in the figure, is moved along the circular path of radius 40 cm from C to D. If the difference in potential energy due to movement of q_3 from C to D is given by $\frac{q_3 K}{4\pi\epsilon_0}$, the value of K is:



[07 April, 2025 (Shift-I)]

- (a) $8q_2$ (d) $6q_2$
 (b) $8q_1$ (c) $6q_1$
2. The electrostatic potential on the surface of uniformly charged spherical shell of radius $R = 10$ cm is 120 V. The potential at the centre of shell, at a distance $r = 5$ cm from centre, and at a distance $r = 15$ cm from the centre of the shell respectively, are:

[03 April, 2025 (Shift-I)]

- (a) 120 V, 120 V, 80 V (b) 40 V, 40 V, 80 V
 (c) 0 V, 0 V, 80 V (d) 0 V, 120 V, 40 V

3. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: Work done in moving a test charge between two points inside a uniformly charged spherical shell is zero, no matter which path is chosen.

Reason R: Electrostatic potential inside a uniformly charged spherical shell is constant and is same as that on the surface of the shell.

In the light of the above statements, choose the correct answer from the options given below

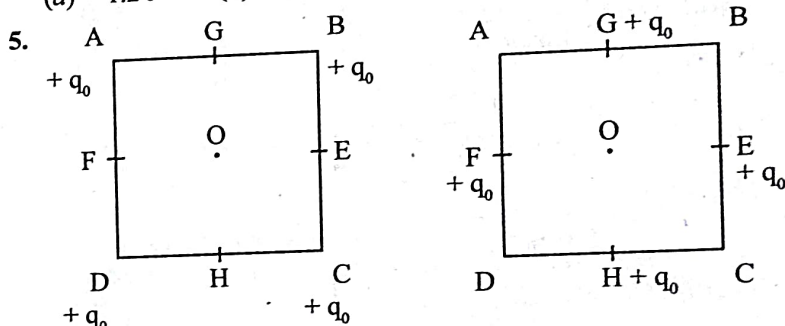
[08 April, 2025 (Shift-II)]

- (a) A is true but R is false
 (b) Both A and R are true and R is the correct explanation of A
 (c) A is false but R is true
 (d) Both A and R are true but R is NOT the correct explanation of A

4. Two charges $7 \mu\text{C}$ and $-4 \mu\text{C}$ are placed at $(-7 \text{ cm}, 0, 0)$ and $(7 \text{ cm}, 0, 0)$ respectively. Given, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$, the electrostatic potential energy of the charge configuration is:

[23 Jan, 2025 (Shift-II)]

- (a) -1.2 J (b) -1.8 J (c) -1.5 J (d) -2.0 J



Configuration (1)

Configuration (2)

In the first configuration (1) as shown in the figure, four identical charges (q_0) are kept at the corners A, B, C and D of square of side length 'a'. In the second configuration (2), the same charges are

shifted to mid points G, E, H and F, of the square. If $K = \frac{1}{4\pi\epsilon_0}$,

the difference between the potential energies of configuration (2) and (1) is given by:

[24 Jan, 2025 (Shift-II)]

- (a) $\frac{Kq_0^2}{a}(4\sqrt{2} - 2)$ (b) $\frac{Kq_0^2}{a}(3 - \sqrt{2})$
 (c) $\frac{Kq_0^2}{a}(3\sqrt{2} - 2)$ (d) $\frac{Kq_0^2}{a}(4 - 2\sqrt{2})$

6. An electric dipole of dipole moment is $6.0 \times 10^{-6} \text{ cm}$ placed in a uniform electric field of $1.5 \times 10^3 \text{ NC}^{-1}$ in such a way that dipole moment is along electric field. The work done in rotating dipole by 180° in this field will be _____ mJ

[8 April, 2023 (Shift-I)]

7. A point charge of $2 \times 10^{-2} \text{ C}$ is moved from P to S in a uniform electric field of 30 N/C directed along positive x-axis. If coordinates of P and S are $(1, 2, 0) \text{ m}$ and $(0, 0, 0) \text{ m}$ respectively, the work done by electric field will be

[29 Jan, 2023 (Shift-II)]

- (a) 1200 mJ (b) 600 mJ (c) -600 mJ (d) -1200 mJ
8. 27 similar drops of mercury are maintained at 10 V each. All these spherical drops combine into a single big drop. The potential energy of the bigger drop is times that of a smaller drop.

[26 Feb, 2021 (Shift-II)]

9. If the potential energy between two molecules is given by $U = -\frac{A}{r^6} + \frac{B}{r^{12}}$, then at equilibrium, separation between molecules, and the potential energy are

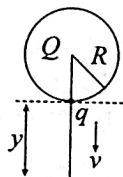
[6 Sep, 2020 (Shift-I)]

$$(a) \left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B} \quad (b) \left(\frac{B}{2A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B}$$

$$(c) \left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{4B} \quad (d) \left(\frac{B}{2A}\right)^{\frac{1}{6}}, 0$$

10. A solid sphere of radius R carries a charge $Q+q$ distributed uniformly over its volume. A very small point like piece of it of mass m gets detached from the bottom of the sphere and falls down vertically under gravity. This piece carries charge q . If it acquires a speed v when it has fallen through a vertical height y (see figure), then (Assume the remaining portion to be spherical).

[5 Sep, 2020 (Shift-I)]



$$(a) v^2 = y \left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g \right]$$

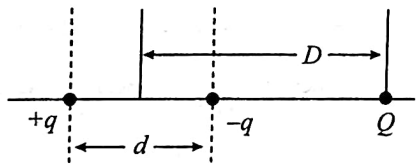
$$(b) v^2 = 2y \left[\frac{qQR}{4\pi\epsilon_0 (R+y)^3 m} + g \right]$$

$$(c) v^2 = y \left[\frac{qQ}{4\pi\epsilon_0 R^2 y m} + g \right]$$

$$(d) v^2 = 2y \left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g \right]$$

11. A system of three charges are placed as shown in the figure. If $D \gg d$, the potential energy of the system is best given by

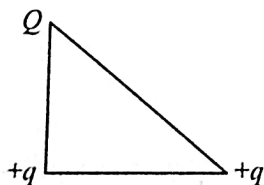
[9 April, 2019 (Shift-I)]



$$(a) \frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right] \quad (b) \frac{1}{4\pi\epsilon_0} \left[+\frac{q^2}{d} + \frac{qQd}{D^2} \right]$$

$$(c) \frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} + \frac{2qQd}{D^2} \right] \quad (d) \frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{2D^2} \right]$$

12. Three charges Q , $+q$ and $+q$ are placed at the vertices of a right-angle isosceles triangles as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is [11 Jan, 2019 (Shift-I)]



$$(a) \frac{-\sqrt{2}q}{\sqrt{2}+1} \quad (b) +q \quad (c) -2q \quad (d) \frac{-q}{1+\sqrt{2}}$$

13. In free space, a particle A of charge $1 \mu\text{C}$ is held fixed at a point P . Another particle B of the same charge and mass $4 \times 10^{-6} \text{ kg}$ is kept at a distance of 1 mm from P . If B is released, then its velocity at a distance of 9 mm from P is

$$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right] \quad [10 \text{ April, 2019 (Shift-II)}]$$

$$(a) 5 \times 10^2 \text{ m/s} \quad (b) 2.0 \times 10^3 \text{ m/s}$$

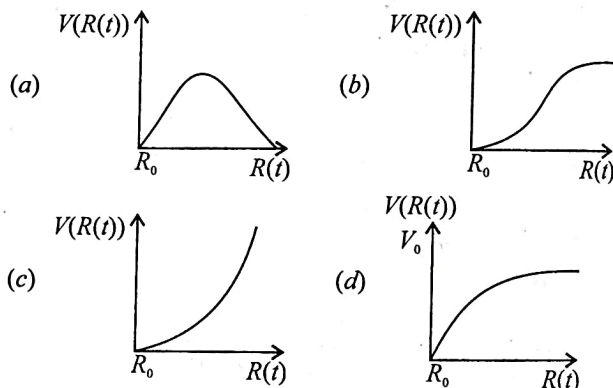
$$(c) 10.0 \text{ m/s} \quad (d) 3.0 \times 10^4 \text{ m/s}$$

14. Four equal point charges Q each are placed in the xy plane at $(0, 2)$, $(4, 2)$, $(4, -2)$ and $(0, -2)$. The work required to put a fifth charge Q at the origin of the coordinate system will be [10 Jan, 2019 (Shift-II)]

$$(a) \frac{Q^2}{2\sqrt{2}\pi\epsilon_0} \quad (b) \frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{3}} \right)$$

$$(c) \frac{Q^2}{4\pi\epsilon_0} \quad (d) \frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{5}} \right)$$

15. There is a uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed $V(R(t))$ of the distribution as a function of its instantaneous radius $R(t)$ is [12 Jan, 2019 (Shift-I)]



16. An electric dipole of dipole moment is $6.0 \times 10^{-6} \text{ cm}$ placed in a uniform electric field of $1.5 \times 10^3 \text{ NC}^{-1}$ in such a way that dipole moment is along electric field. The work done in rotating dipole by 180° in this field will be ____ mJ [8 April, 2023 (Shift-I)]

Electric Potential

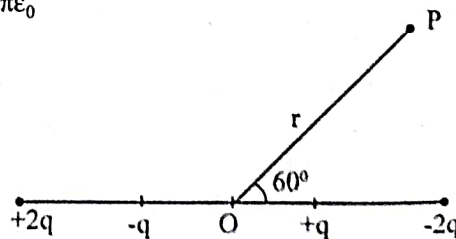
17. An electric charge $10^{-6} \mu\text{C}$ is placed at origin $(0, 0)$ m of X - Y coordinate system. Two points P and Q are situated at $(\sqrt{3}, \sqrt{3})$ m and $(\sqrt{6}, 0)$ m respectively. The potential difference between the points P and Q will be: [27 Jan, 2024 (Shift-I)]

$$(a) \sqrt{3} \text{ V} \quad (b) \sqrt{6} \text{ V} \quad (c) 0 \text{ V} \quad (d) 3 \text{ V}$$

18. The distance between charges $+q$ and $-q$ is $2l$ and between $+2q$ and $-2q$ is $4l$. The electrostatic potential at point P at a distance r from centre O is $-\alpha \left[\frac{ql}{r^2} \right] \times 10^9 \text{ V}$ where the value of α is ____

$$\left(\text{Use } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right)$$

[31 Jan, 2024 (Shift-II)]



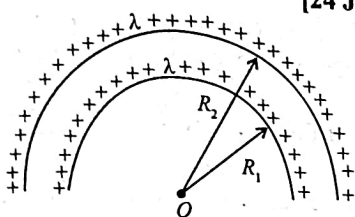
[31 Jan, 2024 (Shift-II)]

19. The electrostatic potential due to an electric dipole at a distance ' r ' varies as: [30 Jan, 2024 (Shift-I)]

- (a) r (b) $\frac{1}{r^2}$ (c) $\frac{1}{r^3}$ (d) $\frac{1}{r}$

20. At the centre of a half ring of radius $R = 10$ cm and linear charge density 4 nC m^{-1} , the potential is $x\pi V$. The value of x is [09 April, 2024 (Shift-I)]

21. The electric potential at the centre of two concentric half rings of radii R_1 and R_2 , having same linear charge density λ is [24 Jan, 2023 (Shift-II)]



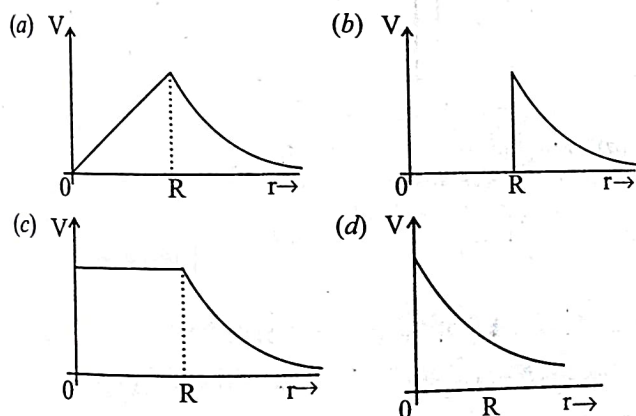
- (a) $\frac{2\lambda}{\epsilon_0}$ (b) $\frac{\lambda}{2\epsilon_0}$ (c) $\frac{\lambda}{4\epsilon_0}$ (d) $\frac{\lambda}{\epsilon_0}$

22. Electric potential at a point 'P' due to a point charge of 5×10^{-9} C is 50 V. The distance of 'P' from the point charge is:

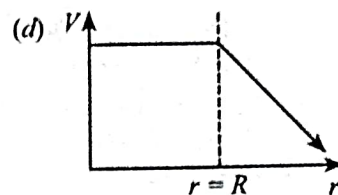
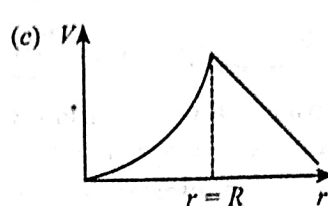
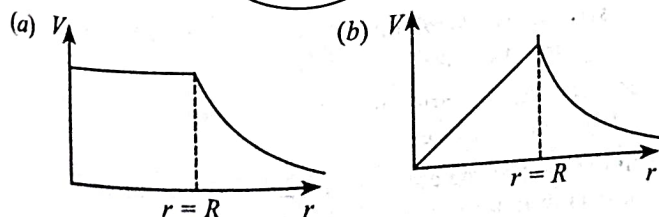
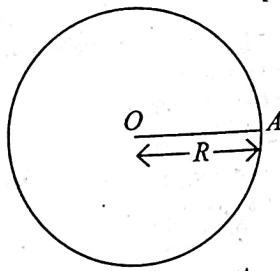
(Assume, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$) [8 April, 2023 (Shift-II)]

- (a) 3 cm (b) 9 cm (c) 90 cm (d) 20.9 cm

23. Which of the following graph correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the centre? [31 Jan 2023 (Shift-I)]



24. For a uniformly charged thin spherical shell, the electric potential (V) radially away from the center (O) of shell can be graphically represented as [6 April, 2023 (Shift-I)]



25. Three concentric spherical metallic shells X , Y and Z of radius a , b and c respectively [$a < b < c$] have surface charge densities σ , $-\sigma$ and σ , respectively. The shells X and Z are at same potential. If the radii of X & Y are 2 cm and 3 cm, respectively. The radius of shell Z is ____ cm. [10 April, 2023 (Shift-I)]

26. 64 identical drops each charged upto potential of 10mV are combined to form a bigger drop. The potential of the bigger drop will be ____ mV. [12 April, 2023 (Shift-I)]

27. 27 identical drops are charged at 22V each. They combine to form a bigger drop. The potential of the bigger drop will be ____ V. [25 June, 2022 (Shift-II)]

28. Two isolated metallic solid spheres of radii R and $2R$ are charged such that both have same charge density σ . The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ' . The ratio $\frac{\sigma'}{\sigma}$ is: [30 Jan, 2023 (Shift-I)]

- (a) $\frac{9}{4}$ (b) $\frac{4}{3}$ (c) $\frac{5}{3}$ (d) $\frac{5}{6}$

29. Sixty four conducting drops each of radius 0.02 m and each carrying a charge of $5\mu\text{C}$ are combined to form a bigger drop. The ratio of surface density of bigger drop to the smaller drop will be [26 June, 2022 (Shift-II)]

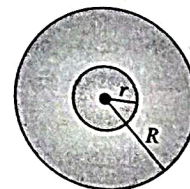
- (a) 1 : 4 (b) 4 : 1 (c) 1 : 8 (d) 8 : 1

30. The two thin coaxial rings, each of radius ' a ' and having charges $+Q$ and $-Q$ respectively are separated by a distance of ' s '. The potential difference between the centres of the two rings is: [26 Aug, 2021 (Shift-II)]

- (a) $\frac{Q}{2\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{\sqrt{s^2 + a^2}} \right]$ (b) $\frac{Q}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{\sqrt{s^2 + a^2}} \right]$
(c) $\frac{Q}{2\pi\epsilon_0} \left[\frac{1}{a} + \frac{1}{\sqrt{s^2 + a^2}} \right]$ (d) $\frac{Q}{4\pi\epsilon_0} \left[\frac{1}{a} + \frac{1}{\sqrt{s^2 + a^2}} \right]$

31. 512 identical drops of mercury are charged to a potential of 2V each. The drops are joined to form a single drop. The potential of this drop is ____ V. [25 Feb, 2021 (Shift-I)]

32. A charge Q is distributed over two concentric conducting thin spherical shells radii r and R ($R > r$). If the surface charge densities on the two shells are equal, the electric potential at the common center is [2 Sep, 2020 (Shift-II)]



- (a) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{(R^2 + r^2)} Q$ (b) $\frac{1}{4\pi\epsilon_0} \frac{(R+2r)Q}{2(R^2 + r^2)}$
(c) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{2(R^2 + r^2)} Q$ (d) $\frac{1}{4\pi\epsilon_0} \frac{(2R+r)}{(R^2 + r^2)} Q$

33. Concentric metallic hollow spheres of radii R and $4R$ hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference $V(R) - V(4R)$ is [3 Sep, 2020 (Shift-II)]

(a) $\frac{3Q_2}{4\pi\epsilon_0 R}$ (b) $\frac{3Q_1}{16\pi\epsilon_0 R}$ (c) $\frac{Q_2}{4\pi\epsilon_0 R}$ (d) $\frac{3Q_1}{4\pi\epsilon_0 R}$

34. A solid conducting sphere, having a charge Q , is surrounded by an uncharged conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V . If the shell is now given a charge of $-4Q$, the new potential difference between the same two surfaces is

[8 April, 2019 (Shift-I)]

(a) $-2V$ (b) V (c) $2V$ (d) $4V$

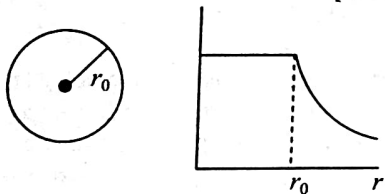
35. A charge Q is distributed over three concentric spherical shell of radii a, b, c ($a < b < c$) such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where $r < a$, would be

[10 Jan, 2019 (Shift-I)]

(a) $\frac{Q(a+b+c)}{4\pi\epsilon_0(a^2+b^2+c^2)}$ (b) $\frac{Q}{4\pi\epsilon_0(a+b+c)}$
(c) $\frac{Q(a^2+b^2+c^2)}{4\pi\epsilon_0(a^3+b^3+c^3)}$ (d) $\frac{Q}{12\pi\epsilon_0} \frac{ab+bc+ca}{abc}$

36. The given graph shows variation (with distance r from center) of

[11 Jan, 2019 (Shift-I)]



- (a) Potential of a uniformly charged spherical shell
(b) Electric field of a uniformly charged sphere
(c) Electric field of uniformly charged spherical shell
(d) Potential of a uniformly charged sphere

37. Considering a group of positive charges, which of the following statements is correct?

[31 Jan, 2023 (Shift-II)]

- (a) Net potential of the system cannot be zero at a point but net electric field can be zero at that point.
(b) Net potential of the system at a point can be zero but net electric field can't be zero at that point.
(c) Both the net potential and the net field can be zero at a point.
(d) Both the net potential and the net electric field cannot be zero at a point.

38. For a charged spherical ball, electrostatic potential inside the ball varies with r as $V = 2ar^2 + b$.

Here, a and b are constants and r is the distance from the center. The volume charge density inside the ball is $-\lambda a \epsilon$. The value of λ is _____.

[29 Jan, 2023 (Shift-II)]

where ϵ = permittivity of medium.

39. Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2 , respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1 (on S_1) / V_2 (on S_2) of the electrostatic potentials on each sphere is

[8 Jan, 2020 (Shift-II)]

(a) $\left(\frac{R_1}{R_2}\right)^2$ (b) $\left(\frac{R_1}{R_2}\right)^3$ (c) $\left(\frac{R_2}{R_1}\right)$ (d) $\frac{R_1}{R_2}$

40. Ten charges are placed on the circumference of a circle of radius R with constant angular separation between successive charges. Alternate charges 1, 3, 5, 7, 9 have charge $(+q)$ each, while 2, 4, 6, 8, 10 have charge $(-q)$ each. The potential V and the electric field E at the center of the circle are respectively. (Take $V = 0$ at infinity)

[5 Sep, 2020 (Shift-II)]

(a) $V = 0; E = 0$ (b) $V = \frac{10q}{4\pi\epsilon_0 R}; E = \frac{10q}{4\pi\epsilon_0 R^2}$
(c) $V = \frac{10q}{4\pi\epsilon_0 R} = E = 0$ (d) $V = 0; E = \frac{10q}{4\pi\epsilon_0 R^2}$

41. A point dipole $\vec{p} = -p_0 \hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the y -axis at a distance d are, respectively

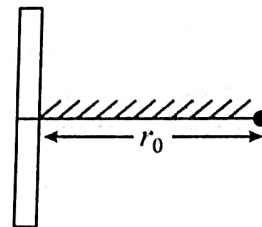
(Take $V = 0$ at infinity)

[12 April, 2019 (Shift-I)]

(a) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$ (b) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$
(c) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$ (d) $0, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$

42. A positive point charge is released from rest at a distance r_0 from a positive line charge with uniform density. The speed (v) of the point charge, as a function of instantaneous distance r from line charge, is proportional to

[8 April, 2019 (Shift-II)]



(a) $v \propto \sqrt{\ln\left(\frac{r}{r_0}\right)}$ (b) $v \propto e^{\frac{r}{r_0}}$
(c) $v \propto \ln\left(\frac{r}{r_0}\right)$ (d) $v \propto \left(\frac{r}{r_0}\right)$

Equipotential Surface, Relation between Field and Potential, Dipole PE

43. Three infinitely long wires with linear charge density λ are placed along the x -axis, y -axis and z -axis respectively. Which of the following denotes an equipotential surface?

[28 Jan, 2025 (Shift-I)]

(a) $(x^2 + y^2)(y^2 + z^2)(z^2 + x^2) = \text{constant}$
(b) $xy + yz + zx = \text{constant}$
(c) $(x + y)(y + z)(z + x) = \text{constant}$
(d) $xyz = \text{constant}$

44. Given below are two statements: one is labelled a Assertion (A) and the other is labelled as Reason (R)

[27 Jan, 2024 (Shift-II)]

Assertion (A): Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R): Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (b) (A) is correct but (R) is not correct
 (c) (A) is not correct but (R) is correct
 (d) Both (A) and (R) are correct and (R) is the correct explanation of (A)

45. If the electric potential at any point (x, y, z) m in space is given by $V = 3x^2$ volt. The electric field at the point $(1, 0, 3)$ m will be:

[29 June, 2022 (Shift-II)]

- (a) 3 Vm^{-1} , directed along positive x -axis.
 (b) 3 Vm^{-1} , directed along negative x -axis.
 (c) 6 Vm^{-1} , directed along positive x -axis.
 (d) 6 Vm^{-1} , directed along negative x -axis.

46. The electric field in a region is given by $\vec{E} = (Ax + B)\hat{i}$, where E is in NC^{-1} and x is in metres. The values of constants are $A = 20$ SI unit and $B = 10$ SI unit. If the potential at $x = 1$ is V_1 and that at $x = -5$ is V_2 , then $V_1 - V_2$ is:

[8 April, 2019 (Shift-II)]

- (a) 180 V (b) -520 V (c) 320 V (d) -48 V

47. Two identical electric point dipoles have dipole moments $\vec{p}_1 = p\hat{i}$ and $\vec{p}_2 = -p\hat{i}$ are held on the x axis at distance 'a' from each other. When released, they move along the x -axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is:

[6 Sep, 2020 (Shift-II)]

- (a) $\frac{p}{a} \sqrt{\frac{1}{\pi\epsilon_0 ma}}$ (b) $\frac{p}{a} \sqrt{\frac{1}{2\pi\epsilon_0 ma}}$
 (c) $\frac{p}{a} \sqrt{\frac{2}{\pi\epsilon_0 ma}}$ (d) $\frac{p}{a} \sqrt{\frac{3}{2\pi\epsilon_0 ma}}$

48. An electric field of 1000 V/m is applied to an electric dipole at angle of 45° . The value of electric dipole moment is 10^{-29} C.m . What is the potential energy of the electric dipole? [11 Jan, 2019 (Shift-II)]

- (a) $-20 \times 10^{18} \text{ J}$ (b) $-7 \times 10^{-27} \text{ J}$
 (c) $-10 \times 10^{-29} \text{ J}$ (d) $-9 \times 10^{-20} \text{ J}$

49. A point dipole $\vec{p} = -p_0\hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the y -axis at a distance d are, respectively

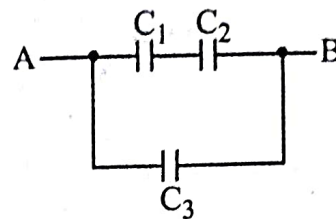
(Take $V = 0$ at infinity)

[12 April, 2019 (Shift-I)]

- (a) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$ (b) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$
 (c) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$ (d) $0, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$

Capacitance, Parallel Plate Capacitor, Dielectric

50. Three parallel plate capacitors C_1, C_2 and C_3 each of capacitance $5 \mu\text{F}$ are connected as shown in figure. The effective capacitance between points A and B, when the space between the parallel plates of C_1 capacitor is filled with a dielectric medium having dielectric constant of 4, is :

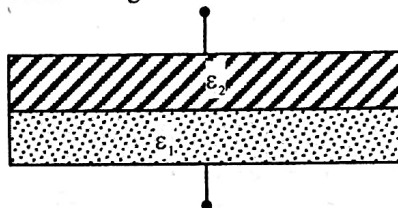


[04 April, 2025 (Shift-II)]

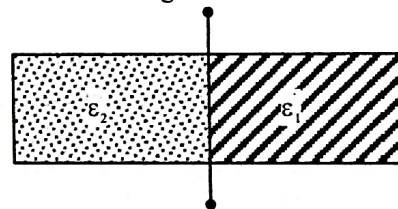
- (a) $22.5 \mu\text{F}$ (b) $7.5 \mu\text{F}$ (c) $9 \mu\text{F}$ (d) $30 \mu\text{F}$

51. A parallel plate capacitor is filled equally (half) with two dielectrics of dielectric constants ϵ_1 and ϵ_2 , as shown in figures. The distance between the plates is d and area of each plate is A . If capacitance in first configuration and second configuration are C_1 and C_2 respectively, then $\frac{C_1}{C_2}$ is:

First Configuration



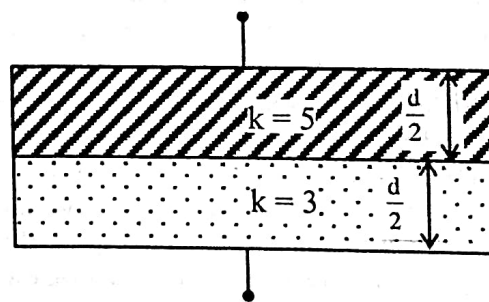
Second Configuration



[03 April, 2025 (Shift-I)]

- (a) $\frac{\epsilon_1 \epsilon_2^2}{(\epsilon_1 + \epsilon_2)^2}$ (b) $\frac{4\epsilon_1 \epsilon_2}{(\epsilon_1 + \epsilon_2)^2}$ (c) $\frac{\epsilon_1 \epsilon_2}{\epsilon_1 + \epsilon_2}$ (d) $\frac{\epsilon_0(\epsilon_1 + \epsilon_2)}{2}$

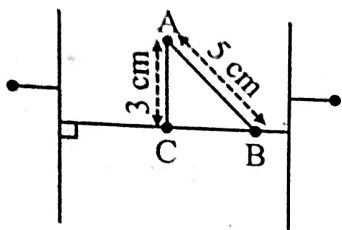
52.



Space between the plates of a parallel plate capacitor of plate area 4 cm^2 and separation of (d) 1.77 mm , is filled with uniform dielectric materials with dielectric constants (3 and 5) as shown in figure. Another capacitor of capacitance 7.5 pF is connected in parallel with it. The effective capacitance of this combination is _____ pF. (Given $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$)

[08 April, 2025 (Shift-II)]

53. Two large plane parallel conducting plates are kept 10 cm apart as shown in figure. The potential difference between them is V . The potential difference between the points A and B (shown in the figure) is:



[02 April, 2025 (Shift-II)]

- (a) $\frac{1}{4} V$ (b) $\frac{2}{5} V$ (c) $\frac{3}{4} V$ (d) $1 V$

54. A parallel plate capacitor has charge $5 \times 10^{-6} C$. A dielectric slab is inserted between the plates and almost fills the space between the plates. If the induced charge on one face of the slab is $4 \times 10^{-6} C$ then the dielectric constant of the slab is _____.
55. A parallel plate capacitor was made with two rectangular plates, each with a length of $l = 3 \text{ cm}$ and breadth of $b = 1 \text{ cm}$. The distance between the plates is $3 \mu\text{m}$. Out of the following, which are the ways to increase the capacitance by a factor of 10?
- A. $l = 30 \text{ cm}$, $b = 1 \text{ cm}$, $d = 1 \mu\text{m}$
 B. $l = 3 \text{ cm}$, $b = 1 \text{ cm}$, $d = 30 \mu\text{m}$
 C. $l = 6 \text{ cm}$, $b = 5 \text{ cm}$, $d = 3 \mu\text{m}$
 D. $l = 1 \text{ cm}$, $b = 1 \text{ cm}$, $d = 10 \mu\text{m}$
 E. $l = 5 \text{ cm}$, $b = 2 \text{ cm}$, $d = 1 \mu\text{m}$

Choose the correct answer from the options given below:

[24 Jan, 2025 (Shift-I)]

- (a) C only (b) B and D only
 (c) A only (d) C and E only
56. A parallel plate capacitor with plate separation 5 mm is charged up by a battery. It is found that on introducing a dielectric sheet of thickness 2 mm , while keeping the battery connections intact, the capacitor draws 25% more charge from the battery than before. The dielectric constant of the sheet is _____ [31 Jan, 2024 (Shift-I)]
57. A parallel plate capacitor of capacitance 12.5 pF is charged by a battery connected between its plates to potential difference of 12.0 V . The battery is now disconnected and a dielectric slab ($\epsilon_r = 6$) is inserted between the plates. The change in its potential energy after inserting the dielectric slab is _____ $\times 10^{-12} \text{ J}$.

[04 April, 2024 (Shift-II)]

58. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): Two metallic spheres are charged to the same potential. One of them is hollow and another is solid, and both have the same radii. Solid sphere will have lower charge than the hollow one.

Reason (R): Capacitance of metallic spheres depend on the radii of spheres.

In the light of the above statements, choose the correct answer from the options given below.

[1 Feb, 2023 (Shift-II)]

- (a) A is false but R is true
 (b) Both A and R are true and R is the correct explanation of A
 (c) A is true but R is false
 (d) Both A and R are true but R is not the correct explanation of A

59. A capacitor of capacitance C is charged to a potential V . The flux of the electric field through a closed surface enclosing the positive plate of the capacitor is: [11 April, 2023 (Shift-II)]

- (a) $\frac{CV}{2\epsilon_0}$ (b) $\frac{2CV}{\epsilon_0}$ (c) $\frac{CV}{\epsilon_0}$ (d) Zero

60. A parallel plate capacitor with air between the plate has a capacitance of 15 pF . The separation between the plate becomes twice and the space between them is filled with a medium of dielectric constant 3.5. Then the capacitance becomes $\frac{x}{4} \text{ pF}$. The value of x is

[24 Jan, 2023 (Shift-II)]

61. A parallel plate capacitor has plate area 40 cm^2 and plate separation 2 mm . The space between the plates is filled with a dielectric medium of thickness 1 mm and dielectric constant 5. The new capacitance of the system is:

[25 Jan, 2023 (Shift-I)]

- (a) $24\epsilon_0 F$ (b) $\frac{3}{10}\epsilon_0 F$ (c) $\frac{10}{3}\epsilon_0 F$ (d) $10\epsilon_0 F$

62. A capacitor has capacitance $5 \mu\text{F}$ when its parallel plates are separated by air medium of thickness d . A slab of a material of dielectric constant 1.5 having area equal to that of plates but thickness $\frac{d}{2}$ is inserted between the plates. Capacitance of the capacitor in the presence of slab will be _____ μF .

[25 Jan, 2023 (Shift-II)]

63. Two identical thin metal plates has charge q_1 and q_2 respectively such that $q_1 > q_2$. The plates were brought close to each other to form a parallel plate capacitor of capacitance C . The potential difference between them is:

[29 July, 2022 (Shift-II)]

- (a) $\frac{(q_1 + q_2)}{C}$ (b) $\frac{(q_1 - q_2)}{C}$ (c) $\frac{(q_1 - q_2)}{2C}$ (d) $\frac{2(q_1 - q_2)}{C}$

64. Capacitance of an isolated conducting sphere of radius R_1 becomes n times when it is enclosed by a concentric conducting sphere of radius R_2 connected to earth. The ratio of their radii $\left(\frac{R_2}{R_1}\right)$ is:

[25 July, 2022 (Shift-II)]

- (a) $\frac{n}{n-1}$ (b) $\frac{2n}{2n+1}$ (c) $\frac{n+1}{n}$ (d) $\frac{2n+1}{2n}$

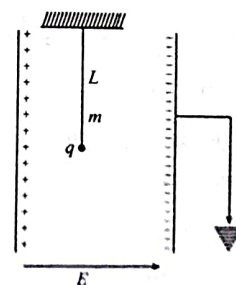
65. A force of 10 N acts on a charged particle placed between two plates of a charged capacitor. If one plate of capacitor is removed, then the force acting on that particle will be. [27 June, 2022 (Shift-I)]

- (a) 5 N (b) 10 N (c) 20 N (d) Zero

66. An electron with kinetic energy K_1 enters between parallel plates of a capacitor at an angle ' α ' with the plates. It leaves the plates at angle ' β ' with kinetic energy K_2 . Then the ratio of kinetic energies $K_1 : K_2$ will be: [25 Feb, 2021 (Shift-I)]

- (a) $\frac{\sin^2 \beta}{\cos^2 \alpha}$ (b) $\frac{\cos \beta}{\cos \alpha}$ (c) $\frac{\cos \beta}{\sin \alpha}$ (d) $\frac{\cos^2 \beta}{\cos^2 \alpha}$

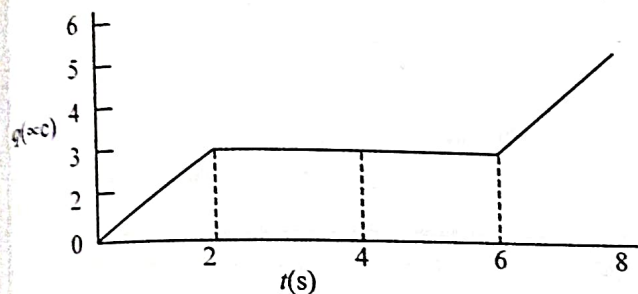
67. A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E , as shown in figure. Its bob has mass m and charge q . The time period of the pendulum is given by [10 April, 2019 (Shift-I)]



(a) $2\pi \frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}$ (b) $2\pi \frac{L}{\sqrt{\left(g + \frac{qE}{m}\right)^2}}$

(c) $2\pi \frac{L}{\sqrt{\left(g - \frac{qE}{m}\right)^2}}$ (d) $2\pi \frac{L}{\sqrt{g^2 - \frac{q^2 E^2}{m^2}}}$

68. The charge on a capacitor plate in a circuit, as a function of time, is shown in the figure: [12 Jan, 2019 (Shift-II)]



What is the value of current at $t = 4$ s?

- (a) zero (b) $3 \propto A$ (c) $2 \propto A$ (d) $63.5 \propto A$

69. A parallel plate capacitor is formed by two plates each of area $30\pi\text{cm}^2$ separated by 1 mm. A material of dielectric strength $3.6 \times 10^7 \text{Vm}^{-1}$ is filled between the plates. If the maximum charge that can be stored on the capacitor without causing any dielectric breakdown is $7 \times 10^{-6} \text{C}$, the value of dielectric constant of the material is: [24 June, 2022 (Shift-I)]

[Use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{Nm}^2\text{C}^{-2}$]

- (a) 64...66 (b) 1.75 (c) 2.25 (d) 2.33

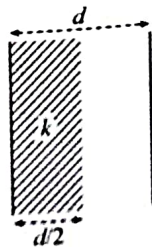
70. A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as:

$\epsilon(x) = \epsilon_0 + kx$, for $\left(0 < x \leq \frac{d}{2}\right)$

$\epsilon(x) = \epsilon_0 + k(d - x)$, for $\left(\frac{d}{2} \leq x \leq d\right)$ [25 July, 2021 (Shift-I)]

- (a) 0 (b) $\frac{kA}{2} \ell n \left(\frac{2\epsilon_0}{2\epsilon_0 - kd} \right)$
- (c) $\frac{kA}{2\ell n \left(\frac{2\epsilon_0 + kd}{2\epsilon_0} \right)}$ (d) $\left(\epsilon_0 + \frac{kd}{2} \right)^{2/k}$

71. In a parallel plate capacitor set up, plate area of capacitor is 2m^2 and the plates are separated by 1 m. If the space between the plates is filled with a dielectric material of thickness 0.5 m and area 2m^2 (See fig) the capacitance of the set-up will be ϵ_0 (Dielectric constant of the material = 3.2) (Round off to the Nearest Integer) [16 March, 2021 (Shift-II)]



72. Voltage rating of a parallel plate capacitor is 500 V. Its dielectric can withstand a maximum electric field of $10^6 \frac{V}{m}$. The plate area is 10^{-4}m^2 . What is the dielectric constant if the capacitance is 15 pF? (given $\epsilon_0 = 8.86 \times 10^{-12} \text{C}^2 / \text{Nm}^2$) [8 April, 2019 (Shift-I)]
- (a) 68...8 (b) 6.2 (c) 4.5 (d) 8.5

73. If q_f is the free charge on the capacitor plates and q_b is the bound charge on the dielectric slab of dielectric constant k placed between the capacitor plates, then bound charge q_b can be expressed as: [25 July, 2021 (Shift-II)]

(a) $q_b = q_f \left(1 + \frac{1}{\sqrt{k}} \right)$ (b) $q_b = q_f \left(1 - \frac{1}{\sqrt{k}} \right)$

(c) $q_b = q_f \left(1 - \frac{1}{k} \right)$ (d) $q_b = q_f \left(1 + \frac{1}{k} \right)$

74. A parallel plate capacitor whose capacitance C is 14 pF is charged by a battery to a potential difference $V = 12 \text{V}$ between its plates. The charging battery is now disconnected and a porcelain plate with $k = 7$ is inserted between the plates, then the plate would oscillate back and forth between the plates with a constant mechanical energy of _____ pJ. (Assume no friction) [17 March, 2021 (Shift-I)]

75. A parallel plate capacitor has plate area 100m^2 and plate separation of 10m. The space between the plates is filled up to a thickness 5m with material of dielectric constant of 10. The resultant capacitance of the system is 'x' pF.

The value of $\epsilon_0 = 8.85 \times 10^{-12} \text{F} \cdot \text{m}^{-1}$

The value of 'x' to the nearest integer is _____.

[18 March, 2021 (Shift-I)]

76. A parallel plate capacitor with plates of area 1m^2 each, are at a separation of 0.1 m. If the electric field between the plates is 100 N/C, the magnitude of charge on each plate is:

[12 Jan, 2019 (Shift-II)]

- (a) $4.85 \times 10^{-10} \text{C}$ (b) $6.85 \times 10^{-10} \text{C}$
- (c) $5.85 \times 10^{-10} \text{C}$ (d) $9.85 \times 10^{-10} \text{C}$

77. A parallel plate capacitor has $1\mu\text{F}$ capacitance. One of its two plates is given $+2\mu\text{C}$ charge and the other plate, $+4\mu\text{C}$ charge. The potential difference developed across the capacitor is:

[8 April, 2019 (Shift-II)]

- (a) 3V (b) 1V (c) 5V (d) 2V

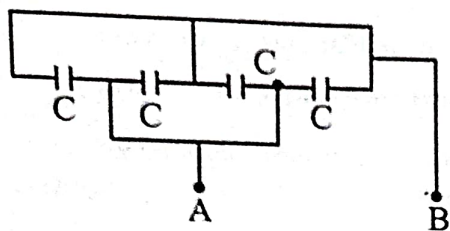
78. The parallel combination of two air filled parallel plate capacitors of capacitance C and nC is connected to a battery of voltage, V . When the capacitors are fully charged, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is

[9 April, 2019 (Shift-II)]

- (a) $\frac{(n+1)V}{(K+n)}$ (b) $\frac{V}{K+n}$ (c) V (d) $\frac{nV}{K+n}$

Combination of Capacitors

79. Four capacitor each of capacitance $16\mu\text{F}$ are connected as shown in the figure. The capacitance between points A and B is: _____ (in μF). [04 April, 2025 (Shift-I)]



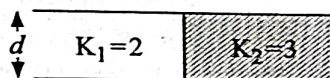
80. Using a battery, a $100 \mu\text{F}$ capacitor is charged to 60 V and then the battery is removed. After that, a second uncharged capacitor is connected to the first capacitor in parallel. If the final voltage across the second capacitor is 20 V , its capacitance is: (in μF)

[03 April, 2025 (Shift-II)]

- (a) 600 (b) 200
(c) 400 (d) 100

81. A capacitor of $10 \mu\text{F}$ capacitance whose plates are separated by 10 mm through air and each plate has area 4 cm^2 is now filled equally with two dielectric media of $K_1 = 2$, $K_2 = 3$ respectively as shown in figure. If new force between the plates is 8 N . The supply voltage is _____ V .

[06 April, 2024 (Shift-II)]



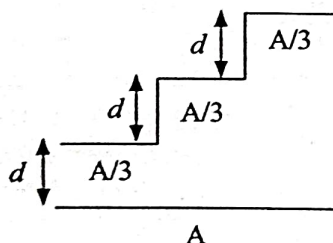
82. A capacitor has air as dielectric medium and two conducting plates of area 12 cm^2 and they are 0.6 cm apart. When a slab of dielectric having area 12 cm^2 and 0.6 cm thickness is inserted between the plates, one of the conducting plates has to be moved by 0.2 cm to keep the capacitance same as in previous case. The dielectric constant of the slab is :

(Given $\epsilon_0 = 8.834 \times 10^{-12} \text{ F/m}$) [08 April, 2024 (Shift-II)]

- (a) 1.50 (b) 1.33 (c) 0.66 (d) 1

83. A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $A/3$ and the height is d , the capacitance of the arrangement is:

[09 April, 2024 (Shift-I)]



- (a) $\frac{11\epsilon_0 A}{18d}$ (b) $\frac{13\epsilon_0 A}{17d}$ (c) $\frac{11\epsilon_0 A}{20d}$ (d) $\frac{11\epsilon_0 A}{20d}$

84. The distance between two plates of a capacitor is d and its capacitance is C_1 , when air is the medium between the plates. If a metal sheet of thickness $\frac{2d}{3}$ and of same area as plate is introduced

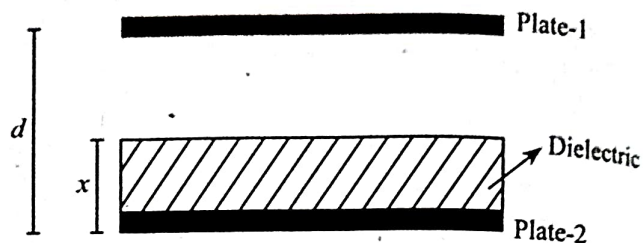
between the plates, the capacitance of the capacitor becomes C_2 .

The ratio $\frac{C_2}{C_1}$ is:

[10 April, 2023 (Shift-II)]

- (a) 2 : 1 (b) 4 : 1 (c) 3 : 1 (d) 1 : 1

85. A parallel plate capacitor with plate area A and plate separation d is filled with a dielectric material of dielectric constant $K = 4$. The thickness of the dielectric material is x , where $x < d$.



Let C_1 and C_2 be the capacitance of the system for $x = \frac{1}{3}d$ and

$x = \frac{2d}{3}$, respectively. If $C_1 = 2\mu\text{F}$ the value of C_2 is _____ μF

[6 April, 2023 (Shift-I)]

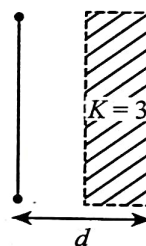
86. Two metallic plates form a parallel plate capacitor. The distance between the plates is ' d '. A metal sheet of thickness $\frac{d}{2}$ and of area equal to area of each plate is introduced between the plates. What will be ratio of the new capacitance to the original capacitance of the capacitor?

[25 June, 2022 (Shift-II)]

- (a) 2 : 1 (b) 1 : 2 (c) 1 : 4 (d) 4 : 1

87. A parallel plate capacitor with plate area A and plate separation $d = 2 \text{ m}$ has a capacitance of $4\mu\text{F}$. The new capacitance of the system if half of the space between them is filled with a dielectric material of dielectric constant $K = 3$ (as shown in figure) will be:

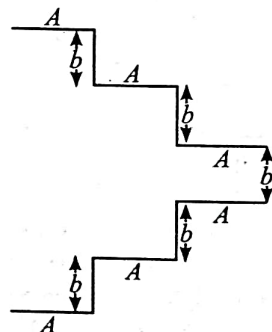
[26 June, 2022 (Shift-II)]



- (a) $2\mu\text{F}$ (b) $32\mu\text{F}$ (c) $6\mu\text{F}$ (d) $8\mu\text{F}$

88. A parallel plate capacitor is made up of stair like structure with a plate area A of each stair and that is connected with a wire of length b , as shown in the figure. The capacitance of the arrangement is $\frac{x\epsilon_0 A}{15b}$, the value of x is _____

[27 June, 2022 (Shift-II)]



89. Consider the combination of 2 capacitors C_1 and C_2 , with $C_1 > C_2$ when connected in parallel, the equivalent capacitance is $\frac{15}{4}$ times the equivalent capacitance of the same connected in series. Calculate the ratio of capacitors, $\frac{C_2}{C_1}$.

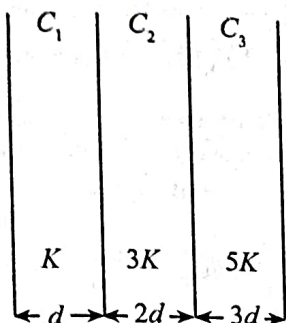
[26 Feb, 2021 (Shift-I)]

- (a) $\frac{111}{80}$ (b) $\frac{15}{4}$ (c) $\frac{29}{15}$ (d) $\frac{15}{11}$

In the reported figure, a capacitor is formed by placing a compound dielectric between the plates of parallel plate capacitor. The expression for the capacity of the said capacitor will be: (Given area of plate = A)

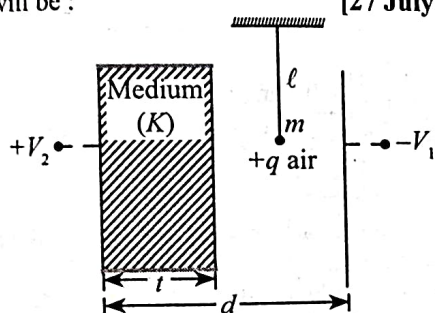
[27 July, 2021 (Shift-I)]

- (a) $\frac{15 K \epsilon_0 A}{34 d}$ (b) $\frac{15 K \epsilon_0 A}{6 d}$ (c) $\frac{25 K \epsilon_0 A}{6 d}$ (d) $\frac{9 K \epsilon_0 A}{6 d}$



A simple pendulum of mass 'm', length 'l' and charge '+q' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be:

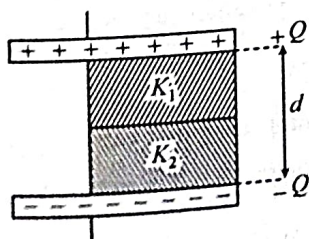
[27 July, 2021 (Shift-II)]



- (a) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_2 (V_2 - V_1)}{(C_1 + C_2)(d - t)} \right]$
 (b) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_1 (V_1 + V_2)}{(C_1 + C_2)(d - 1)} \right]$
 (c) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_1 (V_2 - V_1)}{(C_1 + C_2)(d - t)} \right]$
 (d) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_2 (V_1 + V_2)}{(C_1 + C_2)(d - t)} \right]$

A parallel-plate capacitor with plate area A has separation d between the plates. Two dielectric slabs of dielectric constant K_1 and K_2 of same area $\frac{A}{2}$ and thickness $\frac{d}{2}$ are inserted in the space between the plates. The capacitance of the capacitor will be given by:

[26 Aug, 2021 (Shift-II)]

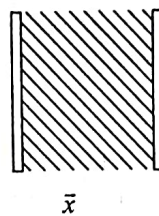


- (a) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{K_1 + K_2}{K_1 K_2} \right)$ (b) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{2(K_1 + K_2)}{K_1 K_2} \right)$
 (c) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{K_1 K_2}{K_1 + K_2} \right)$ (d) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{K_1 K_2}{2(K_1 + K_2)} \right)$

93. A parallel plate capacitor has plates of area A separated by distance 'd' between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x) = k(1 + \alpha x)$ where 'x' is the distance measured from one of the plates.

IF $(\alpha d) \ll 1$, the total capacitance of the system is best given by the expression:

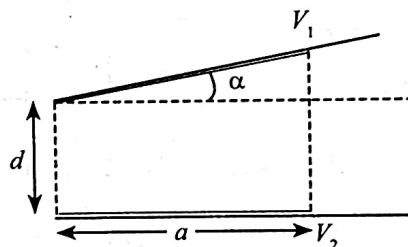
[7 Jan, 2020 (Shift-I)]



- (a) $\frac{AK\epsilon_0}{d} \left(1 + \frac{\alpha d}{2} \right)$ (b) $\frac{A\epsilon_0 K}{d} \left(1 + \frac{\alpha^2 d^2}{2} \right)$
 (c) $\frac{AK\epsilon_0}{d} (1 + \alpha d)$ (d) $\frac{A\epsilon_0 K}{d} \left[1 + \left(\frac{\alpha d}{2} \right)^2 \right]$

94. A capacitor is made of two square plates each of side 'a' making a very small angle α between them, as shown in figure. The capacitance will be close to:

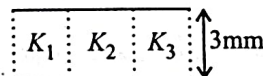
[8 Jan, 2020 (Shift-II)]



- (a) $\frac{\epsilon_0 a^2}{d} \left(1 + \frac{\alpha a}{d} \right)$ (b) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{4d} \right)$
 (c) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{3\alpha a}{2d} \right)$ (d) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d} \right)$

95. A parallel plate capacitor is of area 6 cm² and a separation 3 mm. The gap is filled with three dielectric materials of equal thickness (see figure) with dielectric constants $K_1 = 10$, $K_2 = 12$ and $K_3 = 14$. The dielectric constant of a material which when fully inserted in above capacitor, gives same capacitance would be

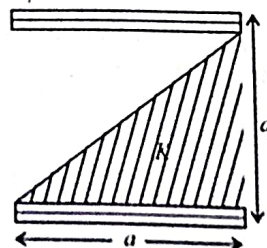
[10 Jan, 2019 (Shift-I)]



- (a) 36 (b) 14 (c) 12 (d) 4

96. A parallel plate capacitor is made of two square plates of side 'a', separated by a distance $d (d \ll a)$. The lower triangular portion is filled with a dielectric of dielectric constant K, as shown in the figure. Capacitance of this capacitor is

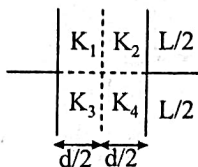
[9 Jan, 2019 (Shift-I)]



- (a) $\frac{K\epsilon_0 a^2}{d(K-1)} \ln K$ (b) $\frac{K\epsilon_0 a^2}{d} \ln K$
 (c) $\frac{K\epsilon_0 a^2}{2d(K+1)}$ (d) $\frac{1}{2} \frac{K\epsilon_0 a^2}{d}$

97. A parallel plate capacitor with square plates is filled with four dielectrics of dielectric constants K_1, K_2, K_3, K_4 arranged as shown in the figure. The effective dielectric constant K will be:

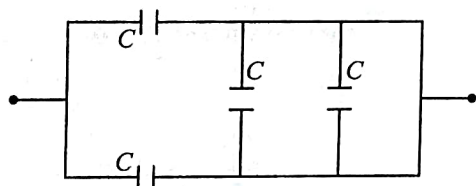
[9 Jan, 2019 (Shift-II)]



- (a) $K = \frac{(K_1 + K_3)(K_2 + K_4)}{K_1 + K_2 + K_3 + K_4}$ (b) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{2(K_1 + K_2 + K_3 + K_4)}$
 (c) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{K_1 + K_2 + K_3 + K_4}$ (d) $K = \frac{(K_1 + K_4)(K_2 + K_3)}{2(K_1 + K_2 + K_3 + K_4)}$

98. The equivalent capacitance of the combination shown is

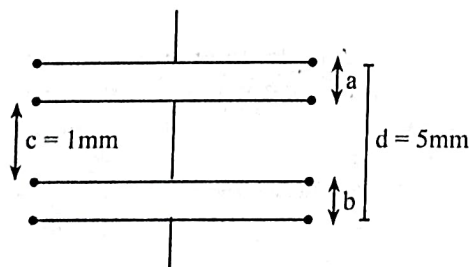
[10 April, 2023 (Shift-I)]



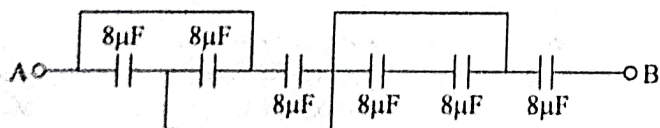
- (a) $\frac{C}{2}$ (b) $4C$ (c) $2C$ (d) $\frac{5}{3}C$

99. As shown in the figure, two parallel plate capacitors having equal plate area of 200 cm^2 are joined in such a way that $a \neq b$. The equivalent capacitance of the combination is $x \epsilon_0 F$. The value of x is

[06 April, 2023 (Shift-II)]



100. The equivalent capacitance between points A and B in below shown figure will be _____ μF . [25 June, 2022 (Shift-I)]

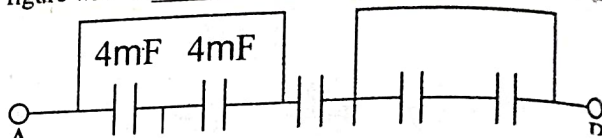


101. Two capacitors, each having capacitance $40 \mu F$ are connected in series. The space between one of the capacitors is filled with dielectric material of constant K such that the equivalence capacitance of the system became $24 \mu F$. The value of K will be:

[28 July, 2022 (Shift-I)]

- (a) 1.5 (b) 2.5 (c) 1.2 (d) 3

102. The equivalent capacitance between points A and B in below shown figure will be _____ μF . [25 June, 2022 (Shift-II)]



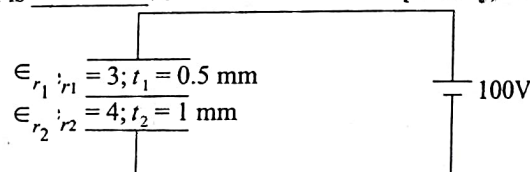
103. A slab of dielectric constant K has the same cross-sectional area as the plates of a parallel plate capacitor and thickness $\frac{3}{4}d$, where d is the separation of the plates. The capacitance of the capacitor when the slab is inserted between the plates will be:

(Given C_0 = capacitance of capacitor with air as medium between plates.)

[28 July, 2022 (Shift-II)]

- (a) $\frac{4KC_0}{3+K}$ (b) $\frac{3KC_0}{3+K}$ (c) $\frac{3+K}{4KC_0}$ (d) $\frac{K}{4+K}$

104. A composite parallel plate capacitor is made up of two different dielectric materials with different thickness (t_1 and t_2) as shown in figure. The two different dielectric materials are separated by a conducting foil F . The voltage of the conducting foil is _____ V. [26 July, 2022 (Shift-I)]



105. For changing the capacitance of a given parallel plate capacitor, a dielectric material of dielectric constant K is used, which has the same area as the plates of the capacitor. The thickness of the dielectric slab is $\frac{3}{4}d$, where ' d ' is the separation between the plates of parallel plate capacitor.

The new capacitance (C') in terms of original capacitance (C_0) is given by the following relation: [16 March, 2021 (Shift-I)]

- (a) $C' = \frac{4+K}{3} C_0$ (b) $C' = \frac{3+K}{4K} C_0$
 (c) $C' = \frac{4}{3+K} C_0$ (d) $C' = \frac{4K}{K+3} C_0$

106. Two capacitors of capacities $2C$ and C are joined in parallel and charge up to potential V . The battery is removed and the capacitor of capacity C is filled completely with a medium of dielectric constant K . The potential difference across the capacitors will now be:

[27 July, 2021 (Shift-I)]

- (a) $\frac{3V}{K}$ (b) $\frac{V}{K}$ (c) $\frac{V}{K+2}$ (d) $\frac{3V}{K+2}$

107. Two equal capacitors are first connected in series and then in parallel. The ratio of the equivalent capacities in the two cases will be:

[24 Feb, 2021 (Shift-I)]

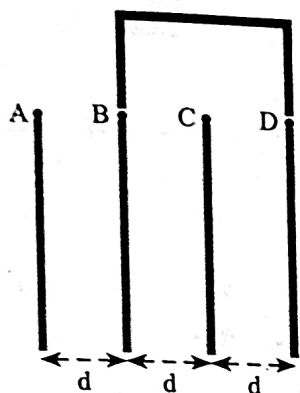
- (a) 4:1 (b) 1:2 (c) 1:4 (d) 2:1

108. Consider the combination of 2 capacitors C_1 and C_2 , with $C_1 > C_2$ when connected in parallel, the equivalent capacitance is $\frac{15}{4}$ times the equivalent capacitance of the same connected in series. Calculate the ratio of capacitors, $\frac{C_2}{C_1}$.

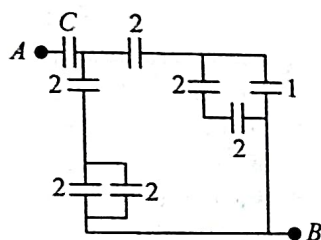
[26 Feb, 2021 (Shift-I)]

- (a) $\frac{111}{80}$ (b) $\frac{15}{4}$ (c) $\frac{29}{15}$ (d) $\frac{15}{11}$

Four identical rectangular plates with length, $\ell = 2$ and breadth, $b = \frac{3}{2}$ cm are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\epsilon_0}{d}$. The value of x is _____
 (Round off to the nearest integer) [17 March, 2021 (Shift-I)]

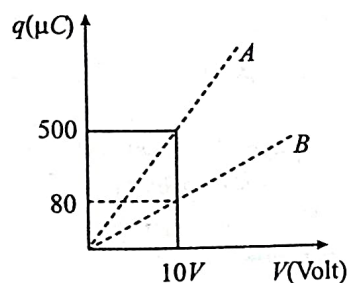


110. In the circuit shown, find C if the effective capacitance of the whole circuit is to be $0.5 \mu\text{F}$. All values in the circuit are in μF . [12 Jan, 2019 (Shift-II)]



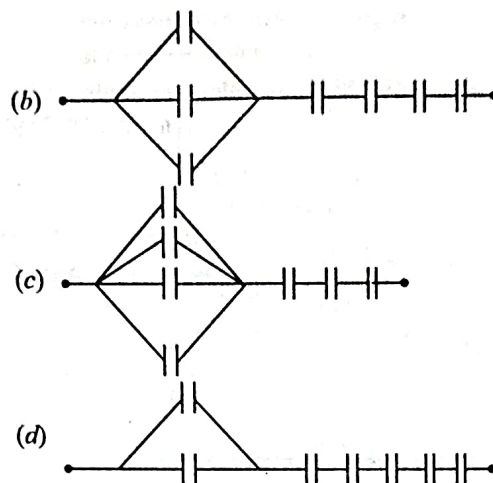
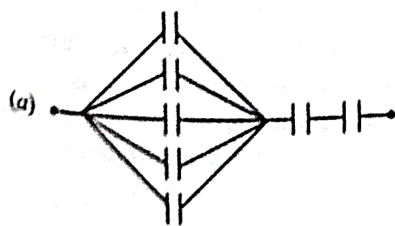
- (a) $\frac{7}{11} \mu\text{F}$ (b) $4 \mu\text{F}$ (c) $\frac{6}{5} \mu\text{F}$ (d) $\frac{7}{10} \mu\text{F}$

111. Figure shows charge (q) versus voltage (V) graph for series and parallel combination of two given capacitors. The capacitances are [10 April, 2019 (Shift-I)]

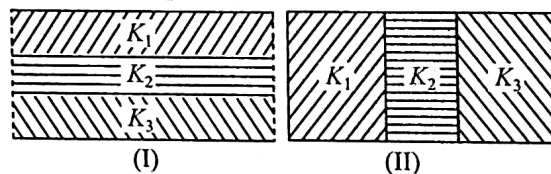


- (a) $40 \mu\text{F}$ and $10 \mu\text{F}$ (b) $20 \mu\text{F}$ and $30 \mu\text{F}$
 (c) $60 \mu\text{F}$ and $40 \mu\text{F}$ (d) $50 \mu\text{F}$ and $30 \mu\text{F}$

112. Seven capacitors, each of the capacitance $2 \mu\text{F}$, are to be connected in a configuration to obtain an effective capacitance of $\left(\frac{6}{13}\right) \mu\text{F}$. Which of the combinations, shown in figures below, will achieve the desired value? [11 Jan, 2019 (Shift-II)]

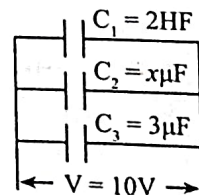


113. Two identical parallel plate capacitors, of capacitance C each, have plates of area A , separated by a distance d . The space between the plates of the two capacitors, is filled with three dielectrics, of equal thickness and dielectric constants K_1 , K_2 and K_3 . The first capacitor is filled as shown in fig. I, and the second one is filled as shown in fig II. If these two modified capacitors are charged by the same potential V , the ratio of the energy stored in the two, would be (E_1 refers to capacitor (I) and E_2 to capacitor (II)) [12 April, 2019 (Shift-I)]

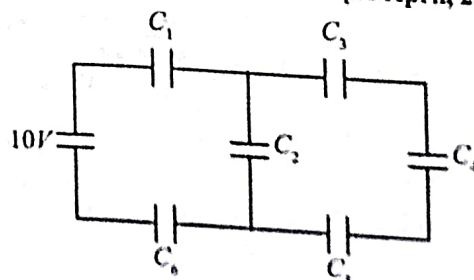


- (a) $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{K_1 K_2 K_3}$
 (b) $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{9 K_1 K_2 K_3}$
 (c) $\frac{E_1}{E_2} = \frac{9 K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$
 (d) $\frac{E_1}{E_2} = \frac{K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$

114. In the given figure the total charge stored in the combination of capacitors is $100 \mu\text{C}$. The value of ' x ' is [15 April, 2023 (Shift-I)]



115. In the given circuit, $C_1 = 2 \mu\text{F}$, $C_2 = 0.2 \mu\text{F}$, $C_3 = 2 \mu\text{F}$, $C_4 = 4 \mu\text{F}$, $C_5 = 2 \mu\text{F}$, $C_6 = 2 \mu\text{F}$, the charge stored on capacitor C_4 is _____ μC . [11 April, 2023 (Shift-II)]



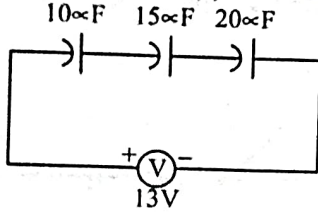
116. The total charge on the system of capacitors $C_1 = 1 \mu F$, $C_2 = 2 \mu F$, $C_3 = 4 \mu F$ and $C_4 = 3 \mu F$ connected in parallel is:
(Assume a battery of 20V is connected to the combination)

[26 July, 2022 (Shift-I)]

- (a) $200 \mu C$ (b) $200 C$ (c) $10 \mu C$ (d) $10 C$
117. The charge on capacitor of capacitance $15 \mu F$ in the figure given below is:

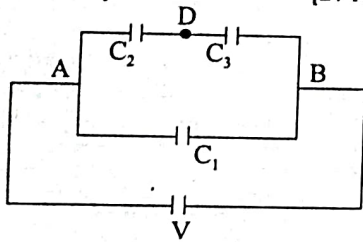
[26 June, 2022 (Shift-II)]

- (a) $60 \mu C$ (b) $130 \mu C$ (c) $260 \mu C$ (d) $585 \mu C$



118. Three capacitors $C_1 = 2 \mu F$, $C_2 = 6 \mu F$ and $C_3 = 12 \mu F$ are connected as shown in figure. Find the ratio of the charges on capacitors C_1 , C_2 and C_3 respectively:

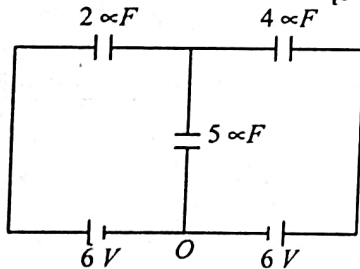
[27 Aug, 2021 (Shift-II)]



- (a) 2:1:1 (b) 3:4:4 (c) 2:3:3 (d) 1:2:2

119. In the circuit shown, charge on the $5 \mu F$ capacitor is:

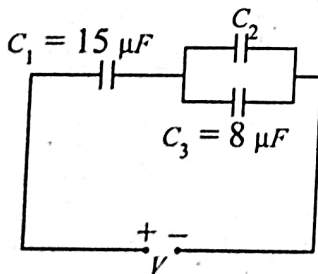
[5 Sep, 2020 (Shift-II)]



- (a) $16.36 \mu C$ (b) $5.45 \mu C$ (c) $10.90 \mu C$ (d) $18.00 \mu C$

120. In the circuit shown in the figure, the total charge is $750 \mu C$ and the voltage across capacitor C_2 is 20 V. Then the charge on capacitor C_2 is

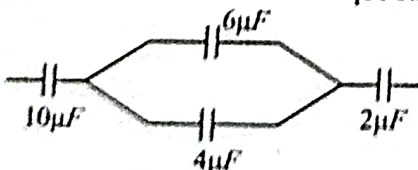
[3 Sep, 2020 (Shift-I)]



- (a) $160 \mu C$ (b) $650 \mu C$ (c) $590 \mu C$ (d) $450 \mu C$

121. In the figure shown below, the charge on the left plate of the $10 \mu F$ capacitor is $-30 \mu C$. The charge on the right plate of the $6 \mu F$ capacitor is

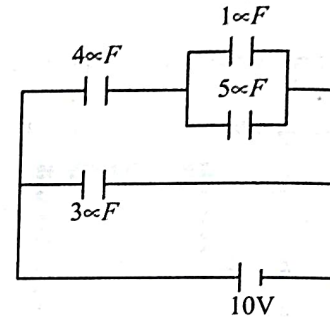
[11 Jan, 2019 (Shift-I)]



- (a) $+18 \mu C$ (b) $-12 \mu C$ (c) $+12 \mu C$ (d) $-18 \mu C$

122. In the given circuit, the charge on $4 \mu F$ capacitor will be:

[12 April, 2019 (Shift-I)]



- (a) $119.4 \mu C$ (b) $24 \mu C$ (c) $9.6 \mu C$ (d) $5.4 \mu C$

Energy Stored in Capacitors & sharing of charge between capacitors

123. The battery of a mobile phone is rated as 4.2 V, 5800 mAh. How much energy is stored in it when fully charged?

[02 April, 2025 (Shift-I)]

- (a) 43.8 kJ (b) 48.7 kJ
(c) 87.7 kJ (d) 24.4 kJ

124. A parallel-plate capacitor of capacitance $40 \mu F$ is connected to a 100 V power supply. Now the intermediate space between the plates is filled with a dielectric material of dielectric constant $K = 2$. Due to the introduction of dielectric material, the extra charge and the change in the electrostatic energy in the capacitor, respectively, are:

[22 Jan, 2025 (Shift-I)]

- (a) 4 mC and 0.2 J (b) 2 mC and 0.2 J
(c) 8 mC and 2.0 J (d) 2 mC and 0.4 J

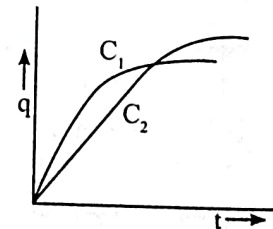
125. Consider a parallel plate capacitor of area A (of each plate) and separation 'd' between the plates. If E is the electric field and ϵ_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is

[24 Jan, 2025 (Shift-I)]

- (a) $\frac{1}{2} \epsilon_0 E^2 A d$ (b) $\epsilon_0 E^2 A d$ (c) $\frac{3}{4} \epsilon_0 E^2 A d$ (d) $\frac{1}{4} \epsilon_0 E^2 A d$

126. Two capacitors C_1 and C_2 are connected in parallel to a battery. Charge-time graph is shown below for the two capacitors. The energy stored with them are U_1 and U_2 , respectively. Which of the given statements is true?

[28 Jan, 2025 (Shift-I)]



- (a) $C_1 > C_2$, $U_1 < U_2$ (b) $C_1 > C_2$, $U_1 > U_2$
(c) $C_2 > C_1$, $U_2 > U_1$ (d) $C_2 > C_1$, $U_2 < U_1$

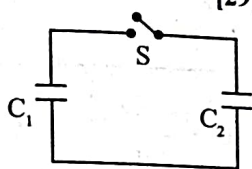
127. A parallel plate capacitor of capacitance $1 \mu F$ is charged to a potential difference of 20 V. The distance between plates is $1 \mu m$. The energy density between plates of capacitor is

[28 Jan, 2025 (Shift-I)]

- (a) $2 \times 10^3 J/m^3$ (b) $1.8 \times 10^3 J/m^3$
(c) $2 \times 10^4 J/m^3$ (d) $1.8 \times 10^4 J/m^3$

A capacitor, $C_1 = 6 \mu\text{F}$ is charged to a potential difference of $V_0 = 5\text{V}$ using a 5V battery. The battery is removed and another capacitor, $C_2 = 12 \mu\text{F}$ is inserted in place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the charges (q_1 and q_2) on the capacitors C_1 and C_2 when equilibrium condition is reached.

[29 Jan, 2025 (Shift-II)]



- (a) $q_1 = 20 \mu\text{C}$, $q_2 = 10 \mu\text{C}$ (b) $q_1 = 15 \mu\text{C}$, $q_2 = 30 \mu\text{C}$
(c) $q_1 = 10 \mu\text{C}$, $q_2 = 20 \mu\text{C}$ (d) $q_1 = 30 \mu\text{C}$, $q_2 = 15 \mu\text{C}$

A capacitor of capacitance C and potential V has energy E . It is connected to another capacitor of capacitance $2C$ and potential $2V$.

Then the loss of energy is $\frac{x}{3} E$, where x is _____.

[30 Jan, 2024 (Shift-I)]

Two identical capacitors have same capacitance C . One of them is charged to the potential V and other to the potential $2V$. The negative ends of both are connected together. When the positive ends are also joined together, the decrease in energy of the combined system is:

[1 Feb, 2024 (Shift-I)]

- (a) $\frac{1}{4} CV^2$ (b) $2 CV^2$ (c) $\frac{1}{2} CV^2$ (d) $\frac{3}{4} CV^2$

Three capacitors of capacitances $25 \mu\text{F}$, $30 \mu\text{F}$ and $45 \mu\text{F}$ are connected in parallel to a supply of 100 V. Energy stored in the above combination is E . When these capacitors are connected in series to the same supply, the stored energy is $\frac{9}{x} E$. The value of x

is _____.

[05 April, 2024 (Shift-I)]

Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is:

[08 April, 2024 (Shift-I)]

- (a) $\sqrt{ab}$ (b) ab (c) a/b (d) b/a

Two parallel plate capacitors C_1 and C_2 each having capacitance of $10 \mu\text{F}$ are individually charged by a 100 V D.C. source. Capacitor C_1 is kept connected to the source and a dielectric slab is inserted between its plates. Capacitor C_2 is disconnected from the source and then a dielectric slab is inserted in it. Afterwards the capacitors are also disconnected from the source and the two capacitors are finally connected in parallel combination. The common potential of the combination will be _____ V. (Assuming Dielectric constant = 10)

[31 Jan, 2023 (Shift-II)]

A parallel plate capacitor of capacitance $200 \mu\text{F}$ is connected to a battery of 200 V. A dielectric slab of dielectric constant 2 is now inserted into the space between plates of capacitor while the battery remain connected. The change in the electrostatic energy in the capacitor will be _____ J.

[31 Aug, 2021 (Shift-II)]

A capacitor of capacitance 50 pF is charged by 100 V source. It is then connected to another uncharged identical capacitor. Electrostatic energy loss in the process is _____ nJ.

[27 June, 2022 (Shift-I)]

136. Effective capacitance of parallel combination of two capacitors C_1 and C_2 is $10 \mu\text{F}$. When these capacitor are individually connected to a voltage source of 1 V, the energy stored in the capacitor C_2 is 4 times that of C_1 . If these capacitors are connected in series, their effective capacitance will be [8 Jan, 2020 (Shift-I)]
(a) $126 \dots 2 \mu\text{F}$ (b) $8.4 \mu\text{F}$ (c) $1.6 \mu\text{F}$ (d) $4.2 \mu\text{F}$

137. A parallel plate capacitor of capacitance 2 F is charged to a potential V . The energy stored in the capacitor is E_1 . The capacitor is now connected to another uncharged identical capacitor in parallel combination. The energy stored in the combination is E_2 . The ratio E_2/E_1 is: [11 April, 2023 (Shift-I)]
(a) $2 : 1$ (b) $1 : 2$ (c) $1 : 4$ (d) $2 : 3$

138. A capacitor of capacitance $900 \mu\text{F}$ is charged by a 100V battery. The capacitor is disconnected from the battery and connected to another uncharged identical capacitor such that one plate of uncharged capacitor connected to positive plate and another plate of uncharged capacitor connected to negative plate of the charged capacitor. The loss of energy in this process is measured as $x \times 10^{-2} \text{ J}$. The value of x is _____ [30 Jan, 2023 (Shift-I)]

139. A 600 pF capacitor is charged by 200 V supply. It is then disconnected from the supply and is connected to another uncharged 600 pF capacitor. Electrostatic energy lost in the process is _____ J.

[8 April, 2023 (Shift-II)]

140. A capacitor of capacitance 50 pF is charged by 100 V source. It is then connected to another uncharged identical capacitor. Electrostatic energy loss in the process is _____ nJ.

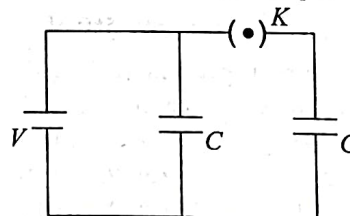
[27 June, 2022 (Shift-I)]

141. If the charge on a capacitor is increased by $2C$, the energy stored in it increases by 44%. The original charge on the capacitor is (in C)

[24 June, 2022 (Shift-II)]

- (a) 10 (b) 20 (c) 30 (d) 40
142. A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key 'K' is closed, the total energy stored across the combination is E_1 . Now key 'K' is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2 . The ratio E_1/E_2 will be:

[26 July, 2022 (Shift-II)]



- (a) $\frac{1}{10}$ (b) $\frac{2}{5}$ (c) $\frac{5}{13}$ (d) $\frac{5}{26}$

143. A parallel plate capacitor filled with a medium of dielectric constant 10, is connected across a battery and is charged. The dielectric slab is replaced by another slab of dielectric constant 15. Then the energy of capacitor will: [29 June, 2022 (Shift-I)]

- (a) Increase by 50% (c) Increase by 25%
(b) Decrease by 15% (d) Increases by 33%

144. A parallel plate capacitor with width 4 cm, length 8cm and separation between the plates of 4 mm is connected to a battery of 20 V. A dielectric slab of dielectric constant 5 having length 1 cm, width 4 cm and thickness 4 mm is inserted between the plates of parallel plate capacitor. The electrostatic energy of this system will be _____ $\epsilon_0 \text{ J}$.

(Where ϵ_0 is the permittivity of free space)

[27 July, 2022 (Shift-II)]

145. Two parallel plate capacitors of capacity C and $3C$ are connected in parallel combination and charged to a potential difference 18 V . The battery is then disconnected and the space between the plates of the capacitor of capacity C is completely filled with a material of dielectric constant 9 . The final potential difference across the combination of capacitors will be _____ V.

[25 July, 2022 (Shift-II)]

146. A capacitor C is fully charged with voltage V_0 . After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\frac{C}{2}$. The energy loss in the process after

the charge is distributed between the two capacitor is:

[4 Sep, 2020 (Shift-II)]

- (a) $\frac{1}{6}CV_0^2$ (b) $\frac{1}{2}CV_0^2$ (c) $\frac{1}{4}CV_0^2$ (d) $\frac{1}{3}CV_0^2$

147. A parallel plate capacitor has plate of length ' ℓ ', width ' w ' and separation of plates is ' d '. It is connected to a battery of emf V . A dielectric slab of the same thickness ' d ' and of dielectric constant $K=4$ is being, inserted between the plates of the capacitor. At what length of the slab inside plates, will be energy stored in the capacitor be two times the initial energy stored? [5 Sep, 2020 (Shift-II)]

- (a) $2\ell/3$ (b) $\ell/2$ (c) $\ell/4$ (d) $\ell/3$

148. Two capacitors of capacitances C and $2C$ are charged to potential differences V and $2V$, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is:

[5 Sep, 2020 (Shift-I)]

- (a) $\frac{3}{2}CV^2$ (b) $\frac{9}{2}CV^2$ (c) $\frac{25}{6}CV^2$ (d) zero

149. A parallel plate capacitor having capacitance 12 pF is charged by a battery to a potential difference of 10 V between its plates. The charging battery is now disconnected and a porcelain slab of dielectric constant 6.5 is slipped between the plates. The work done by the capacitor on the slab is:

[10 Jan, 2019 (Shift-II)]

- (a) 692 pJ (b) 508 pJ (c) 560 pJ (d) 600 pJ

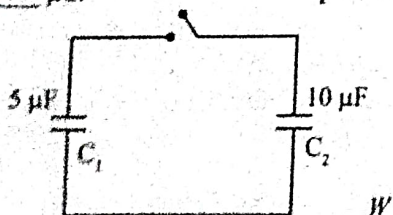
150. A capacitor with capacitance $5\text{ }\mu\text{F}$ is charged to $5\text{ }\mu\text{C}$. If the plates are pulled apart to reduce the capacitance to $2\text{ }\mu\text{F}$, how much work is done?

[9 April, 2019 (Shift-I)]

- (a) $3.75 \times 10^{-6}\text{ J}$ (b) $2.55 \times 10^{-6}\text{ J}$
(c) $142.25 \times 10^{-6}\text{ J}$ (d) $2.16 \times 10^{-6}\text{ J}$

151. A capacitor C_1 of capacitance $5\text{ }\mu\text{F}$ is charged to a potential of 30 V using a battery. The battery is then removed and the charged capacitor is connected to an uncharged capacitor C_2 of capacitance $10\text{ }\mu\text{F}$ as shown in figure. When the switch is closed charge flows between the capacitors. At equilibrium, the charge on the capacitor C_2 is _____ μC .

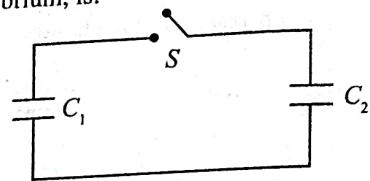
[28 June, 2022 (Shift-II)]



152. Two capacitors having C_1 and C_2 respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential

difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S . The amount of charge on the capacitor C_2 after equilibrium, is:

[26 June, 2022 (Shift-I)]



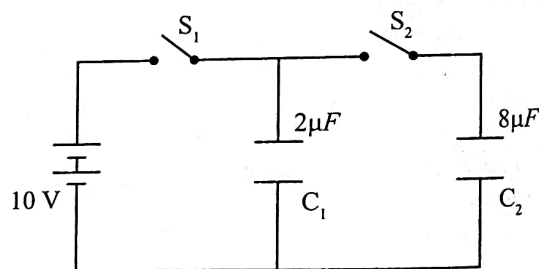
- (a) $\frac{C_1 C_2}{(C_1 + C_2)}V$ (b) $\frac{(C_1 + C_2)}{C_1 C_2}V$ (c) $(C_1 + C_2)V$ (d) $(C_1 - C_2)V$

153. A $10\text{ }\mu\text{F}$ capacitor is fully charged to a potential difference of 50 V . After removing the source voltage, it is connected to an uncharged capacitor in parallel. Now potential difference across them becomes 20 V . The capacitance of the second capacitor is:

- (a) $15\text{ }\mu\text{F}$ (b) $20\text{ }\mu\text{F}$ (c) $10\text{ }\mu\text{F}$ (d) $30\text{ }\mu\text{F}$

154. A $2\text{ }\mu\text{F}$ capacitor C_1 is first charged to a potential difference of 10 V using a battery. Then the battery is removed and the capacitor is connected to an uncharged capacitor C_2 of $8\text{ }\mu\text{F}$. The charge in C_2 on equilibrium condition is _____ μC . (Round off to the Nearest integer)

[17 March, 2021 (Shift-II)]



155. Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have $12\text{ }\mu\text{C}$ and $-3\text{ }\mu\text{C}$ charges, respectively, and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively:

[3 Sep, 2020 (Shift-I)]

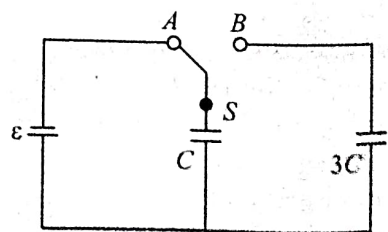
- (a) $+4.5\text{ }\mu\text{C}$ and $-4.5\text{ }\mu\text{C}$ (b) $148.5\text{ }\mu\text{C}$ on both
(c) $6\text{ }\mu\text{C}$ and $3\text{ }\mu\text{C}$ (d) $3\text{ }\mu\text{C}$ and $6\text{ }\mu\text{C}$

156. A $5\text{ }\mu\text{F}$ capacitor is charged fully by a 220 V supply. It is then disconnected from the supply and is connected in series to another uncharged $2.5\text{ }\mu\text{F}$ capacitor. If the energy change during the charge redistribution is $\frac{X}{100}\text{ J}$ then value of X to the nearest integer is:

[2 Sep, 2020 (Shift-II)]

157. In the figure shown, after the switch ' S ' is turned from position ' A ' to position ' B ', the energy dissipated in the circuit in terms of capacitance ' C ' and total charge ' Q ' is _____

[12 Jan, 2019 (Shift-I)]



- (a) $\frac{3Q^2}{8C}$ (b) $\frac{1Q^2}{8C}$ (c) $\frac{5Q^2}{8C}$ (d) $\frac{3Q^2}{4C}$

ANSWER KEY

1. (a)	2. (a)	3. (b)	4. (b)	5. (c)	6. [18]	7. (c)	8. [243]	9. (c)	10. (d)
11. (a)	12. (a)	13. (b)	14. (d)	15. (d)	16. [18]	17. (c)	18. [27]	19. (b)	20. [36]
21. (b)	22. (c)	23. (c)	24. (a)	25. [5]	26. [160]	27. [198]	28. (d)	29. (b)	30. (a)
31. [128]	32. (a)	33. (b)	34. (b)	35. (a)	36. (a)	37. (a)	38. [12]	39. (a)	40. (a)
41. (c)	42. (a)	43. (a)	44. (d)	45. (d)	46. (a)	47. (b)	48. (b)	49. (c)	50. (c)
51. (b)	52. [15]	53. (b)	54. [5]	55. (d)	56. [2]	57. [750]	58. (a)	59. (c)	60. [105]
61. (c)	62. [6]	63. (c)	64. (a)	65. (a)	66. (d)	67. (a)	68. (a)	69. (d)	70. (c)
71. [3]	72. (d)	73. (c)	74. [864]	75. [161]	76. (c)	77. (b)	78. (a)	79. [64]	80. (b)
81. (80)	82. (a)	83. (a)	84. (c)	85. [3]	86. (a)	87. (c)	88. [23]	89. (*)	90. (a)
91. (d)	92. (c)	93. (a)	94. (d)	95. (c)	96. (a)	97. (*)	98. (c)	99. [5]	100. [6]
101. (a)	102. [12]	103. (a)	104. [60]	105. (d)	106. (d)	107. (c)	108. (*)	109. [2]	110. (a)
111. (a)	112. (b)	113. (c)	114. [5]	115. [4]	116. (a)	117. (a)	118. (d)	119. (a)	120. (c)
121. (a)	122. (b)	123. (c)	124. (a)	125. (a)	126. (c)	127. (b)	128. (c)	129. [2]	130. (a)
131. [86]	132. (c)	133. [55]	134. [4]	135. [125]	136. (c)	137. (b)	138. [225]	139. [6]	140. [125]
141. (a)	142. (c)	143. (a)	144. [240]	145. [6]	146. (a)	147. (d)	148. (a)	149. (b)	150. (a)
151. [100]	152. (a)	153. (a)	154. [16]	155. (c)	156. [4]	157. (a)			